

Predicting Post-Harvest Freshness Using Simulated Hyperspectral Signatures

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INTRODUCTION, RESEARCH GAP & RESEARCH QUESTION

Fresh fruits and vegetables begin to change internally soon after harvest, even when they still look fresh on the outside.¹ Because many freshness checks depend on appearance alone, early spoilage can be missed, contributing to food waste.⁴

The FreshID project shows that hyperspectral imaging, AI, and biochemical analysis can detect hidden changes in produce quality before visible deterioration appears.^{1,2,5}

Research gap: There is a need for low-cost, theory-based models that can identify which freshness indicators are most informative under varying storage and maturity conditions without full laboratory deployment.^{4,5}



Research question: Can simulated hyperspectral and biochemical features be used to rank the most useful indicators of post-harvest freshness before visible spoilage appears?

OBJECTIVES

- 1 Study how post-harvest senescence changes produce quality over time.
- 2 Identify the most promising freshness indicators from existing literature.
- 3 Build a simulation-based model to predict freshness trends.
- 4 Evaluate which variables are most informative for early freshness detection.

ACKNOWLEDGEMENTS

I would like to thank Dr. Juan F. Arratia and his organization, the Scientific Caribbean Foundation, Inc., for mentorship and guidance, Professor Komla Folly for support and academic direction, the University of Cape Town for providing the environment to pursue this research, and the University of Florida–FreshID team for their research and inspiration.



ABSTRACT



Post-harvest produce quality decreases after harvest because internal biochemical and structural changes begin before visible spoilage appears. This project explores a simulation-based framework for predicting freshness using hyperspectral signatures and key biochemical indicators. The goal is to identify the most informative variables for freshness detection without needing lab equipment.

MATERIALS & METHODS

This project uses a simulation-based approach guided by published research and FreshID-style concepts. The methodology follows a five-stage process.



1. Literature Review
Comprehensive review of post-harvest senescence, hyperspectral imaging, and biochemical markers.



2. Variable Selection
Identify key indicators: storage time, maturity, firmness, glucosinolate level, and spectral response.



3. Simulation Model
Build a computational model in Python to simulate freshness decline and spectral responses.



4. Feature Ranking
Rank the most informative variables using statistical and machine learning techniques.



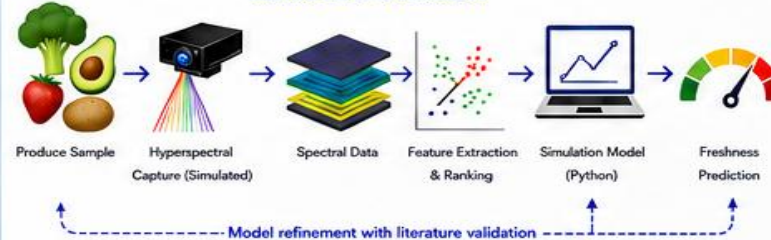
5. Theoretical Validation
Compare simulated results with published research findings to verify expected trends.

Model Equation (Freshness Score)

$$F_{\text{score}} = w_1(\text{Spectral Response}) + w_2(\text{Firmness}) + w_3(\text{Storage Time}) + w_4(\text{Biochemical Marker}) + b$$

Where: $w_1 - w_4$ are weights learned through feature ranking, and b is a bias term. The model predicts freshness on a scale from 0 (spoiled) to 1 (very fresh).

WORKFLOW OVERVIEW



TOOLS USED

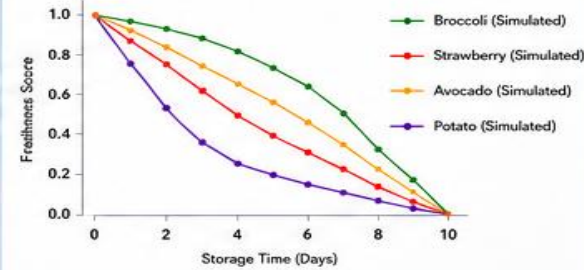


KEY INDICATORS CONSIDERED

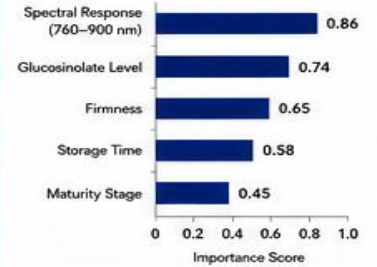


RESULTS (SIMULATED)

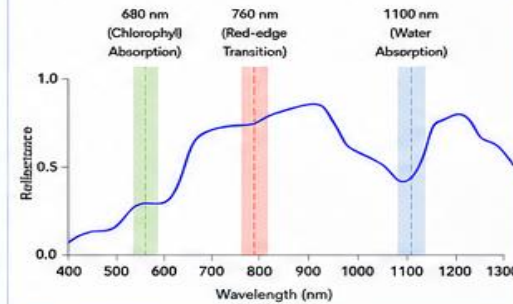
1. Simulated Freshness Decline Over Time



2. Top Influential Variables (Simulation)



3. Key Spectral Regions Linked to Freshness Indicators



Region	Wavelength (nm)	Associated Change	Why It Matters
Green (Chlorophyll)	~680	Chlorophyll degradation	Indicates early aging and loss of green pigments
Red-edge	~760	Cell structure changes	Strong indicator of senescence and maturity shift
NIR (Water Absorption)	~1100	Water content decline	Related to firmness loss and freshness

KEY TAKEAWAYS

- ✓ Freshness declines nonlinearly, with faster drop after mid-storage.
- ✓ Red-edge (760–900 nm) and NIR bands carry the most information.
- ✓ Biochemical markers (e.g., glucosinolates in broccoli) change before visible spoilage.²
- ✓ A small set of variables can effectively predict freshness in the simulated model.

FUTURE WORK

- Validate the simulation model using real hyperspectral datasets.
- Expand the model to more crops and storage conditions.
- Integrate multi-omics data (RNA, proteins, metabolites) for improved accuracy.
- Explore hardware optimization for low-cost, portable freshness detection devices.

REFERENCES

1. University of Florida. (n.d.). FreshID Project – Machine learning for the molecular evolution of fresh produce quality. <http://portal.nifa.usda.gov/web/crisprojectpages/1026004-fact-freshid-machine-learning-for-the-molecular-evaluation-of-fresh-produce-quality.html>
2. Li, S., et al. (2023). Evaluation of postharvest senescence of broccoli via hyperspectral imaging and biochemical analysis. *Postharvest Biology and Technology*, 199, 112180.
3. Zhang, L., et al. (2022). Digitization of broccoli freshness integrating external color and mass loss with multiple internal qualities. *Frontiers in Nutrition*, 9, 1001831.
4. Gowen, A. A., et al. (2007). Hyperspectral imaging—An emerging process analytical tool for food quality and safety control. *Trends in Food Science & Technology*, 18(1–2), 590–598.
5. Feng, L., et al. (2021). Hyperspectral imaging for fruit and vegetable quality evaluation: A review. *Food Chemistry*, 356, 129712.

Abstract + Introduction

ABSTRACT

Post-harvest produce quality decreases because internal biochemical and structural changes begin before visible spoilage appears. This project explores a simulation-based framework for predicting freshness using hyperspectral signatures and key biochemical indicators. The goal is to identify the most informative variables for freshness detection without needing lab equipment.

INTRODUCTION + RESEARCH GAP

Fresh fruits and vegetables often look acceptable on the outside while quality is already declining inside. The FreshID project shows that hyperspectral imaging, AI, and biochemical analysis can detect hidden freshness changes before visible deterioration appears[1,2,4].

Research gap: there is still room for a low-cost, theory-driven model that tests which freshness indicators are most informative under different storage and maturity conditions without laboratory instrumentation[1,4].

Research question: can simulated hyperspectral and biochemical features be used to rank the most useful indicators of post-harvest freshness before visible spoilage appears?

PROJECT CONTEXT

Focus: A first-year, simulation-based extension of the FreshID concept.

Crop examples: broccoli, avocado, strawberry, and potato.

Main idea: track hidden freshness changes before visible spoilage.

Hyperspectral

AI / ML

Simulation

SIMPLE MODEL IDEA

Freshness Score = $w_1(\text{Spectral Response}) + w_2(\text{Firmness}) + w_3(\text{Storage Time}) + w_4(\text{Biochemical Marker})$

Model structure informed by literature-driven freshness indicators. This keeps the project theoretical, but still lets the model rank which variables matter most.

Core takeaway: the project asks whether a simple computational model can capture early freshness loss before the produce looks spoiled.

Materials and Method

RESEARCH AIM

Identify which freshness indicators are most useful in a simulation-based model.

WHY THIS APPROACH

- Intentionally simulation based so realistic for a first-year project
- Does not require lab equipment
- Still connects directly to the FreshID research direction[1]
- Can later be tested on real data

METHOD OVERVIEW



VARIABLES + MODEL

Storage time • Maturity stage • Firmness • Spectral response • Glucosinolate level [2,3]

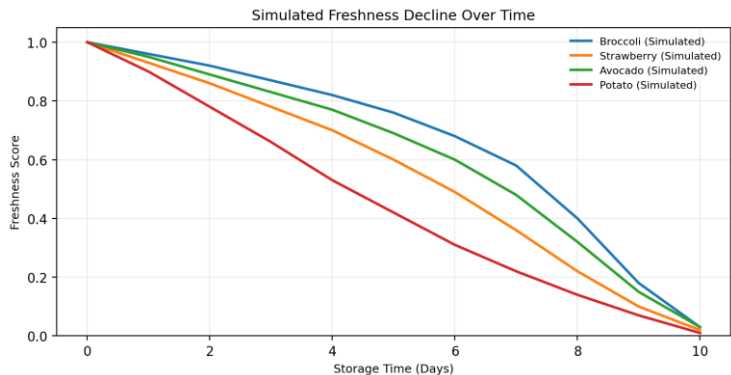
Simulation logic: combine the variables above to estimate a freshness score and rank feature importance.

METHOD STEPS

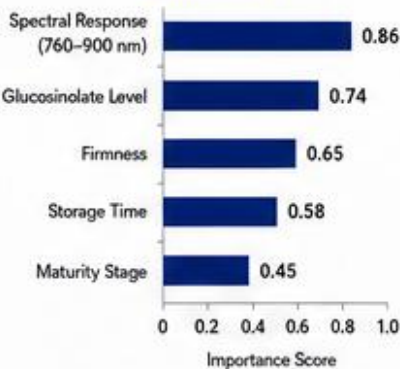
1. Literature review → 2. Variable selection → 3. Python simulation → 4. Feature ranking → 5. Theoretical validation

Results and Future Work

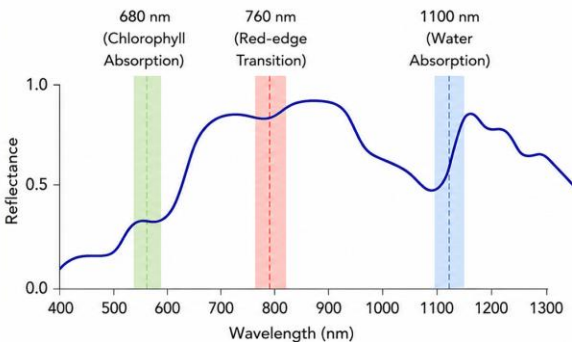
RESULTS (SIMULATED)



2. Top Influential Variables (Simulation)



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KEY TAKEAWAYS

- Freshness declines nonlinearly over storage time[2,3]
- Biochemical change may appear before visible spoilage[2,3]
- Red-edge and near-infrared features are likely the most informative[2,4]
- A small set of variables can still give useful predictions

MODEL INTERPRETATION

This does not replace the full FreshID system. It supports it by showing which inputs should matter first in a low-cost, theory-driven model.

FUTURE WORK

- Test the framework on real hyperspectral data to see how data and results will vary.
- Expand to avocado, strawberry, and potato.
- Improve the model with more variables and storage conditions.
- Use the simulation as a base for future experimental validation.

Important References and Acknowledgements

MOST USEFUL REFERENCES

1. University of Florida / USDA NIFA. FreshID — Machine learning for the molecular evaluation of fresh produce quality.
2. Li et al. (2023). Evaluation of Postharvest Senescence of Broccoli via Hyperspectral Imaging.
3. Zhang et al. (2022). Digitization of Broccoli Freshness Integrating External Color and Mass Loss.
4. Feng et al. (2021). Hyperspectral imaging and freshness detection in fruits and vegetables.
5. Dr Arratia J.F. (2026). Scientific Method, Scientific Abstract & Scientific Poster workshop slides.

ACKNOWLEDGEMENTS

Thank you to Dr. Juan F. Arratia, Professor Komla Folly, the University of Cape Town, and the University of Florida FreshID team for support, guidance, and research inspiration.

PRESENTATION NOTE

This project is intentionally kept simulation-based so it stays realistic for a first-year student while still contributing to the broader freshness-detection research direction.